

Exam Date & Time: 06-Dec-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Deep learning in Computer Vision [CSE-5041]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A researcher is designing a convolutional neural network to classify images in the MNIST dataset. One of the early layers in this CNN applies a linear filter (convolutional filter) to detect edges and textures within the images. Each input image is 28×28 pixels in grayscale. Suppose the researcher wants to use a 3×3 edge-detection filter. The kernel for a simple edge detection filter is given by:

$$K = \begin{pmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{pmatrix} \quad (5)$$
 1. Calculate the output (feature map) size after applying this 3×3 filter to the 28×28 input image, assuming a stride of 1 and "valid" padding (no zero-padding).
What would the output size be if the "same" padding (zero-padding of 1 on each side) was used instead?
- B) Provide an example of how combining SURF with deep learning could enhance performance, particularly in tasks where SURF's robustness to scale and rotation is beneficial. (3)
- C) How does scale invariance improve the robustness of a deep learning model when detecting objects of varying sizes? (2)
- 2) Compare the performance of deep learning-based corner detection and feature matching with traditional methods (e.g., Harris detector + SIFT). Discuss the pros and cons of each approach, focusing on robustness, accuracy, and computational cost. (5)
 - A) Describe the steps to build a Bag of Words model for text data preprocessing in a deep learning pipeline. What challenges might arise with vocabulary size? (3)
 - C) How does the Hough Transform detect lines in an image, and what are the main parameters that need to be adjusted for accurate line detection? (2)
- 3) Multi-layer Perceptrons (MLPs) often struggle with certain challenges when applied to complex datasets, especially in image and sequence data tasks. Discuss the following issues that can arise with MLPs in deep learning, and explain potential solutions or alternative architectures that might address these limitations: High Computational Cost (5)
 - A)

and Overfitting

- B) AlexNet uses dropouts to prevent overfitting during training. Explain how dropout works and why it was critical in AlexNet's architecture. What might happen if dropout layers were removed, particularly when training on smaller datasets? (3)
- C) How do skip connections in ResNet impact the network's ability to learn complex features in deeper layers? (2)
- 4) Given that deconvolution layers use learned kernels for upscaling, how might the random initialization of these kernels contribute to the checkerboard patterns observed in the output images? What other issues could arise from improper initialization? (5)
- A)
- B) What is the purpose of Class Activation Mapping (CAM) in deep learning, and how does it help in visualizing the decision-making process of a Convolutional Neural Network (CNN)? (3)
- C) What is the main objective of the DeepDream algorithm in deep learning, and how does it create surreal, dream-like images? (2)
- 5) What advantages would the Fast RNN bring to improve the speed and efficiency of this real-time speech recognition system which is built using traditional RNN? (5)
- A)
- B) What is the role of max-pooling indices in SegNet, and how do they contribute to better segmentation performance compared to traditional fully convolutional networks (FCNs)? (3)
- C) Consider the traditional RNN update rule and the Fast RNN modification. Write the standard RNN update formula and then provide the Fast RNN update formula. How does the Fast RNN modify the computation of the hidden state to achieve computational efficiency? (2)

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